Cameron DeBough

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EDUCATION

University of Arizona, B.S. Computer Science & Artificial Intelligence, Minor in Mathematics **Expected May 2029**
GPA: 3.87 · Dean's List with Distinction (Fall 2025) · Dean's List (Spring 2026) · 67 Credits Completed
Coursework: Software Development, Discrete Mathematics for CS I-II, Analyzing Data, Web Programming, Calculus I-II, Linear Algebra

EXPERIENCE

Software Engineering Intern, SkillfullyAware / SAAQ **October 2025 - Present**
- Own the production backend of an AI-powered leadership assessment platform used by executive peer organizations (EO, YPO): Python and FastAPI services on Supabase (PostgreSQL), live in production
- Built a streaming LLM analysis pipeline with multi-stage scoring and two-pass generation that eliminated timeouts and truncated output, plus automated DOCX report generation from the analysis results
- Implementing authentication, Stripe payments, and persistent user storage for MVP launch

Founder & Engineer, DeBough Systems (deboughsystems.com) **August 2026 - Present**
- Built and launched a cybersecurity service that helps small businesses with no IT staff find and fix the weak spots attackers target most, with every issue explained in plain English
- Wrote a read-only checkup program that scans each company computer in about 40 seconds (updates, antivirus, firewall, disk encryption) and turns the results into a ranked fix list, a weekly email report, and a printable report for cyber insurers
- Built a free domain scan that checks whether attackers could fake a company's email or impersonate it with look-alike websites; Next.js, TypeScript, Supabase/Postgres, and Stripe, with rate limiting and ownership checks to prevent misuse

RESEARCH EXPERIENCE

Undergraduate Research Assistant, VSI Lab, University of Arizona ECE **September 2025 - Present**
Embodied AI & Accessibility · Advisor: Prof. Eungjoo Lee
- Led a real-robot study of online Gaussian Splatting SLAM (MonoGS) on a TurtleBot 4 against offline structure-from-motion (COLMAP, GLOMAP + 3DGS); online mapping took 2.4 min vs. 4.0 min for the fastest offline pipeline, at 31.0 vs. 29.7 dB PSNR
- Built the ROS 2 pipeline end to end: rosbag2 recording, dataset conversion, SLAM/SfM runs, and ATE, PSNR, SSIM, LPIPS evaluation
- Found the OAK-D depth camera reads 11 to 13% long and corrected it with lidar, cutting SLAM tracking error from 7.4 to 1.2 cm
- Also building NVIDIA Isaac Sim environments, ROS 2 control of LeRobot SO-101 arms, and computer-vision pipelines for Meta Aria 2 smart glasses (hand tracking, gaze data) for ASL accessibility

Research Software Engineer, NewFOS Center, University of Arizona **March 2026 - Present**
New Frontiers of Sound, an NSF Science & Technology Center
- Built Python automation pipelines for nonlinear acoustics experiments (hysteresis sweeps, harmonic SNR analysis, transfer-function extraction) used to model signal distortion in acoustic transducers
- Designed and validated an electrical impedance measurement rig with a full de-embedding pipeline that isolated a 42% cable-parasitic error in a previously reported measurement
- Developed MATLAB signal-analysis pipelines (SVD, dynamic mode decomposition) that produced publication-quality figures for experiments on a three-rod coupled acoustic waveguide

PRESENTATIONS & AWARDS

Poster, Bay Area Robotics Symposium (BARS 2026), UC Berkeley **October 2026**
Online Gaussian Splatting SLAM on a TurtleBot4: Real-Robot Evaluation Against Offline Structure-from-Motion. C. DeBough, W. Koh, E. Lee

Undergraduate Research Travel Grant ($1,000), University of Arizona **Fall 2026**

TECHNICAL PROJECTS

- **Thermal-Tracking Turret:** Raspberry Pi turret that tracks a heat source in real time (I2C thermal array, PWM servos, closed-loop control)
- **APEX QUANT:** event-driven backtesting and paper-trading engine for micro futures with risk controls and an Interactive Brokers layer
- **Synapse Finance:** full-stack financial SaaS with 50+ active users (Flask, React, PostgreSQL, JWT auth, CI/CD)

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, MATLAB, Bash, HTML/CSS
Frameworks: FastAPI, Flask, React, Next.js, Node.js, Tailwind CSS
Robotics & Hardware: ROS 2, SLAM, Gaussian Splatting, COLMAP, Isaac Sim, Raspberry Pi, I2C, PWM servo control, VISA/SCPI
AI & Data: PyTorch, agentic AI (tool calling, RAG), LLM integration, computer vision, NumPy, SciPy, Pandas, signal processing
Tools & Databases: Git, Docker, Linux, PostgreSQL, Supabase, Stripe, CI/CD

LEADERSHIP & ACTIVITIES

Coke Fellow, The Coca-Cola Company, Introships Program (1 of 50 fellows nationwide) **May - July 2026**

Founder & Owner, Durian Car Detailing **May 2023 - Present**
- Built and run a car detailing business from scratch with 200+ jobs completed, handling pricing, marketing, and customer service

University of Arizona AI Club · MLH Global Hack Week · 13+ GitHub projects